

(Non)commutative ball maps

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Theorem[Beurling's theorem in finite codimension]

Let $\mathcal{M}$ be invariant, $\dim \mathcal{M}^\perp = n < \infty$

- ▶ There is a rational function φ , regular across the boundary of $\mathbb{D}$ such that $P_{\mathcal{M}} = M_\varphi M_\varphi^*$ and

$$|\varphi(z)| = 1 \quad \text{whenever} \quad |z| = 1$$

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What we want: finite codimension multiplier-invariant subspaces should be ranges of “rational”, “inner” functions, and maybe more details...

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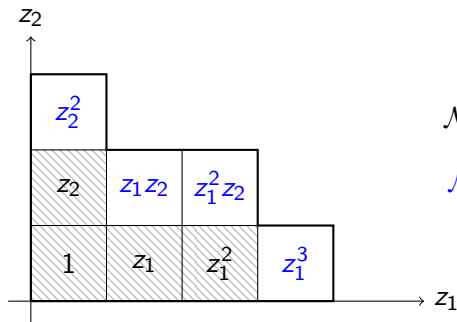
- ▶ say a closed subspace $\mathcal{M} \subset H_g^2$ is invariant if $\varphi \mathcal{M} \subset \mathcal{M}$ for all multipliers (equivalently, just for coordinate functions z_j)

Let $\mathcal{M} \subset H_g^2$ be an invariant subspace. Define

$$\mathcal{M}_\partial = \text{span}\{1, z_j u : u \in \mathcal{M}^\perp\} \ominus \mathcal{M}^\perp$$

The ball codimension of $\mathcal{M}$ is

$$m := \dim \mathcal{M}_\partial$$



$$\mathcal{M}^\perp = \text{span}\{1, z_1, z_2, z_1^2\}$$

$$\mathcal{M}_\partial = \text{span}\{z_2^2, z_1 z_2, z_1^2 z_2, z_1^3\}$$

$$m = 4$$

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- ▶ (ball map) there is a rational multiplier $\varphi : \mathbb{B}^g \rightarrow \mathbb{B}^m$, smooth across the boundary, such that $P_{\mathcal{M}} = M_\varphi M_\varphi^*$ and

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- ▶ (minimality and uniqueness) if there is another multiplier $\psi : \mathbb{B}^g \rightarrow \mathbb{B}^M$ with $P_{\mathcal{M}} = M_\psi M_\psi^*$

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So when $g = 1$, ball codimension is always 1, and φ is a finite Blaschke product

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Theorem 2 (zero-based invariant subspaces) If $\Lambda = \{\lambda_1, \dots, \lambda_n\}$ are distinct points and $\mathcal{M}$ is the space of functions vanishing on Λ , then

$$\dim \mathcal{M}_\partial = n(g - 1) + 1 \quad (\text{the upper bound is attained})$$

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Problem: Give a satisfying account of the “gaps” (compare SCV works of D’Angelo et al.)

hydration break

pause fraîcheur

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- ▶ View F as a function on f -tuples of $n \times n$ matrices
 $Z = (Z_1, \dots, Z_d)$

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where $Z^w = Z_{i_1} Z_{i_2} \cdots Z_{i_n}$ for $w = i_1 i_2 \cdots i_n$

- ▶ View F as a function on g -tuples of $n \times n$ matrices
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- ▶ There is a Beurling-style theorem for L (or R) invariant subspaces $\mathcal{M} \subset \mathcal{F}_g^2$ (Arias/Popescu '95, Davidson/Pitts '98)

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- ▶ choice of free point $\mathbf{Y}$ is unique up to a similarity

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N.B. no “ball codimension” in the free setting (or, it’s always the maximal value $n(g - 1) + 1$), and so no “gap conditions”